THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — MATHS · MEDIUM LEVEL CHAPTER TEST SET 3

Chapter 4: **Expressions using Letter-Numbers** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The nth term of a pattern is $5n + 7$. Its 8th term is: 1

(a) 47 **(b)** 46 **(c)** 48 **(d)** 49

2. A pencil costs ₹x and an eraser costs ₹y. The cost of 4 pencils and 7 erasers is: 1

(a) $28(x + y)$ **(b)** $11xy$ **(c)** $4x + 7y$ **(d)** $4y + 7x$

3. If $a = 5$, the value of $a^2 + 3a$ is: 1

(a) 39 **(b)** 42 **(c)** 40 **(d)** 41

4. If $x = 3$, the value of $\frac{8x+4}{4}$ is: 1

(a) 7 **(b)** 8 **(c)** 6 **(d)** 9

5. The value of $7x + 2y - 4$ for $x = 3$ and $y = 8$ is: 1

(a) 33 **(b)** 35 **(c)** 34 **(d)** 32

6. Simplify $15x + 2y - 9x + 8y$: 1

(a) $24x + 10y$ **(b)** $16xy$ **(c)** $6x + 10y$ **(d)** $6x - 6y$

7. If $x = 10$, then $2(x - 4) + 9$ equals: 1

(a) 22 **(b)** 21 **(c)** 23 **(d)** 20

8. Subtract $6x + 3$ from $11x + 12$: 1

(a) $-5x - 9$ **(b)** $17x + 15$ **(c)** $5x + 9$ **(d)** $5x + 15$

9. The nth term of a pattern is $3n + 9$. Its 15th term is: 1

(a) 54 **(b)** 53 **(c)** 56 **(d)** 55

10. A pencil costs ₹x and an eraser costs ₹y. The cost of 8 pencils and 3 erasers is: 1

(a) $8y + 3x$ **(b)** $11xy$ **(c)** $8x + 3y$ **(d)** $24(x + y)$

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Find the value of $x^2 + 5x - 7$ when $x = 4$. 2

12. Simplify: $5(x + 2) + 8x$. 2

13. Subtract $5x - 4$ from $12x + 9$. 2

14. Find the value of $x^2 + 4x - 3$ when $x = 7$. 2

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SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Two numbers are $12n + 7$ and $5n - 6$. Find their sum and their difference (first minus second), and evaluate both for $n = 3$. 3

16. A row of n triangles made with matchsticks needs $2n + 1$ sticks. How many sticks are needed for 20 triangles and for 30 triangles? 3

17. A cab charges ₹40 plus ₹16 per km. Write the fare for k km. Find the fare for 7 km and for 12 km. 3

OR

Two numbers are $11n + 2$ and $4n - 1$. Find their sum and their difference (first minus second), and evaluate both for $n = 6$. 3

18. A row of n triangles made with matchsticks needs $2n + 1$ sticks. How many sticks are needed for 11 triangles and for 16 triangles? 3

OR

A cab charges ₹55 plus ₹12 per km. Write the fare for k km. Find the fare for 9 km and for 14 km. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. A shopkeeper sells x kg of rice at ₹60 per kg and y kg of wheat at ₹35 per kg. Write an expression for his total sales and find it when $x = 9$ and $y = 14$. 5

OR

Raju is r years old. His mother's age is 3 more than 3 times his age. After 4 years, write the ages of both and the sum of their ages. Evaluate the sum for $r = 14$. 5

20. A shopkeeper sells x kg of rice at ₹44 per kg and y kg of wheat at ₹32 per kg. Write an expression for his total sales and find it when $x = 15$ and $y = 20$. 5

OR

Raju is r years old. His mother's age is 7 more than 3 times his age. After 2 years, write the ages of both and the sum of their ages. Evaluate the sum for $r = 13$. 5

END OF QUESTION PAPER — Total: 40 Marks

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